Sense Together

Individual Project
Service Design



Deaf and Blind individuals face distinct challenges when attending large outdoor music festivals, from navigating crowded spaces to accessing performance information. However, most festivals in China still lack inclusive and differentiated design to support their needs. This project proposes **an accessibility-focused service system that integrates AR smart glasses, haptic bracelet, and improved spatial planning** to enhance mobility, safety, and sensory engagement for people with hearing or visual impairments.

2025/6/12-2025/10/30

Insperation

I often attend music festivals. At the Heliu Music Festival in Xuzhou, I saw sign language interpreters on stage, which made me realize the challenges disabled people face at musical events. How do the hearing impaired experience music? What support do the visually impaired need? In China, disabled audiences still face significant barriers. I aim to design a service framework that enables equal participation in music festivals.



Case Study

Case 1: Rock in Rio 2024 (Brazil)



Time: September 13 ~ 22, 2024

Venue: Rio de Janeiro, Brazil

Situation

As one of Latin America's largest music festivals, it runs for seven days, drawing 100,000 people daily

Accessibility Services

Previously, interpreters worked only on stage, but this year their signing was displayed on the big screens **for the first time in 40 years**

Reserved areas for disabilities at major stages

A "Feel the Sound" area near the stage lets deaf and hard-of-hearing (DHH) visitors experience music **through vibrations**

Case 2: Heliu Music Festival 2025 (China)

Time: April 11 ~ 13, 2025

Venue： Malingshan Scenic Area, Xuzhou, China

Situation

It is China's first music festival to provide accessible sign language interpretation services, it runs for three days

Accessibility Services

Sign language interpreters assist deaf visitors near the stage

Entrances have **accessible lanes** for disabilities

Volunteers from special education programs provide guidance and support



Background Research



Deaf kid who love music

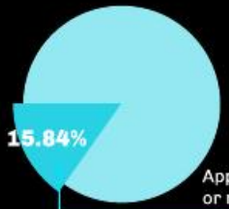
"We **yearn to be part of the larger community** and to be understood and cared for."

"I attended LAY Zhang's concert, which **featured sign language interpretation**—it was a wonderful experience."

"Whether young or old, regardless of background, able-bodied or intellectually impaired **—no one should be excluded because of their differences.**"



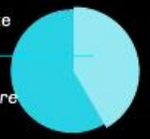
Organizer of a music festival



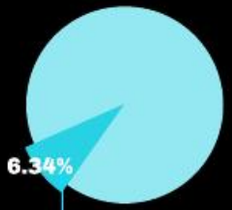
Approximately **206 million** people in China have hearing impairments

Accounting for **15.84%** of the country's total population

Approximately **67.2 million** people have moderate or more severe (significantly impactful) hearing loss



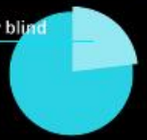
People with mild hearing impairments are largely unfamiliar with sign language



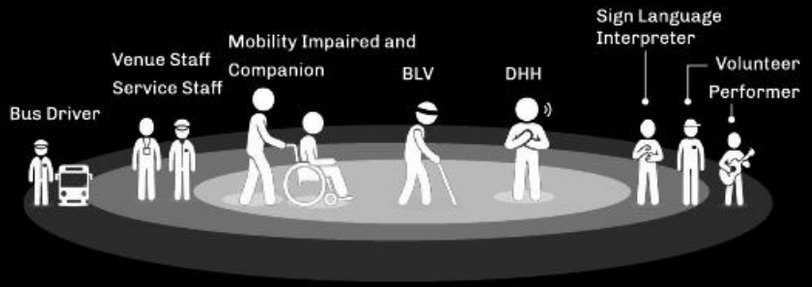
Approximately **17 million** people in China have visual impairments

Accounting for **6.34%** of the national population

approximately **4 million** are totally blind



Stakeholder



Interviews



AZan

Shandong, China **Moderate hearing impairment**

Through the sign language interpreter, I could follow the lyrics, feel the singer's emotions, and fully enjoy the show. Without one, **I'd miss stage talks or audience shouts.**
This inclusivity made us **feel respected** and proved **music transcends hearing.**



Na Liu

Sichuan, China **Mild hearing impairment**

I don't understand sign language. I've checked AR translation glasses, but **they're not sold.** I just need subtitles and lyrics, since **my hearing aids pick up too much background noise.** I can read lips, but **the stage is too far to see clearly.**



Jingyang Lv

Beijing, China **Class I Visual Impairment**

I hope there could be designated viewing areas for blind individuals, preferably with handrails nearby. Although I can navigate independently, music festivals are extremely crowded, making it difficult to use my white cane for pathfinding and maintaining balance.

Insights

For the hearing impaired:

People with mild hearing impairments do not know sign language; they greatly need real-time translation captions.

While hearing-impaired individuals can hear sounds with hearing aids, they often struggle to make out speech clearly, resulting in a suboptimal listening experience.

For the visually impaired:

In crowded places, it is more difficult for visually impaired individuals to use their white canes to navigate and maintain balance.

Market Research



Mozart Festival Orchestra

Location: UK

Product: Vibrotextile™ Haptic Vest System

- Designed for DHH audiences to experience live music through vibration
- Allows users to feel rhythm and subtle nuances of orchestral performances

Coachella Festival

Location: California, USA

- Provided accessibility services for DHH attendees
- ADA wristbands granted access to accessible viewing platforms
- ASL interpreters available at major stages with published schedules
- Offered Assistive Listening Devices (ALDs) like FM systems for enhanced sound



Strengths	Limitations
Inclusive, immersive, enhances emotional connection	Expensive, needs calibration, limited accessibility

Product Research

Haptic Feedback Devices

Woojer Vest Edge (Woojer, 2020)



8 vibration modules provide 360° tactile feedback
Strong bass immersion and multi-platform connection
Bulky, expensive, and uncomfortable for outdoor use

SubPac X1 Wearable (SubPac, 2021)



Lightweight and compact, this model focuses on low-frequency vibration for DJs and concert-goers. It offers portability and long battery life but only provides feedback on the back area and it's costly

AR Smart Glasses

RayNeo X2 (TCL)



Uses waveguide optics and full-color MicroLED, bright enough for outdoor stages
Supports AI translation, live captions, and rhythm visualization
App ecosystem is still developing

XREAL Air 2 + Beam



Compact and comfortable for long use
Displays fixed "floating" visuals (lyrics, rhythm animations)
Short battery life and relies on external devices

Persona



Ying Zhao
26 years old
Hearing-Impaired

Background
Ying Zhao is a deaf festival-goer who relies on real-time subtitles and visual cues. She loves the energy of live events but **often misses lyrics and emotional details**, making her feel partially excluded

Behaviors and Needs
Relies on white cane for navigation and handrails for stability
Prefers having designated viewing areas for safety and comfort

Pain Points
Music festival venues are crowded and difficult to navigate independently
Poor contrast and unclear ground signage make wayfinding harder

Statement
I don't use sign language
I prefer reading subtitles and lyrics while watching performances

Behaviors and Needs
Can read lips but struggles with distance visibility from the stage
Uses AR translation subtitles for understanding lyrics

Pain Points
Background noise reduces clarity of speech and lyrics
Music volume at concerts often too low for hearing aids

Goals
To have customized visual support tools **(e.g., real-time captions, AR lyrics)**
To experience clear sound and visible expressions during performances

Statement
I enjoy attending music festivals but faces mobility challenges in crowded environments
I want to move around the festival independently without relying on or inconveniencing friends

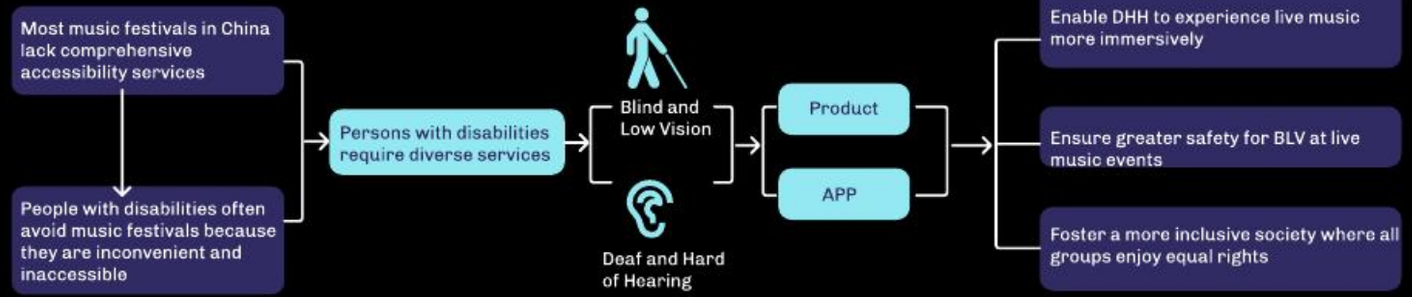
Goals
To enjoy music events safely and independently
To have tactile or auditory navigation aids that enhance spatial awareness



Tianyang Zhou
28 years old
Visually-Impaired

Background
Tianyang Zhou is a BLV music lover who enjoys festivals but struggles to move safely in crowds. He depends on friends' support, and limited accessibility services reduce his independence

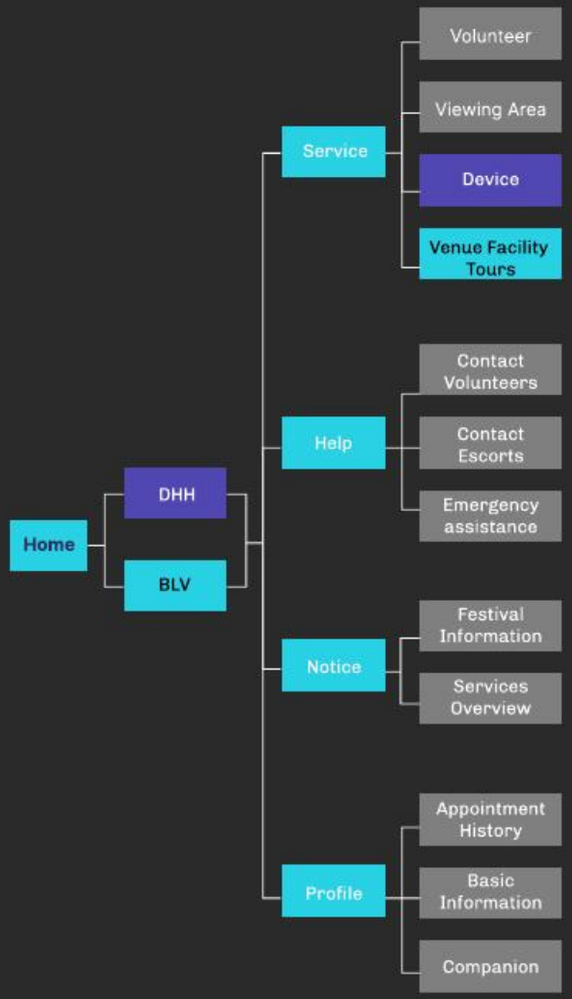
Concept



Survice Blueprint

Phase	Before the Performance					During the performance			After the performance	
User Action *Line of Interaction*	Registration Purchase disability-accessible tickets Submit Service Request	Board at designated locations Take the accessible shuttle bus directly to the venue Verify the identity upon boarding	Enter through the special access point Obtain an identity wristband	Contact Volunteers Pick up assistive devices at the Accessibility Service Center	Service personnel provide guidance on using the product View the product's electronic user manual	Enter the fenced-off viewing area	Enter the priority viewing area Watch the real-time sign language interpretation up close Activate the haptic bracelet and put on the AR smart glasses	Accessible facilities are available during intermission Share the performance experience with fellow audience members around	Return the equipment to the Accessibility Service Center	Exit via special passage Alight at designated stops when riding the accessible shuttle bus
Front Stage *Line of Visibility*	App interface displays information App interface provides accessibility service options	My Identity Information Interface Accessible Performance Shuttle	App QR Code Ticket Verification Interface Service staff distribute identification wristbands	Verify identity with the authentication wristband Confirm Service Requirements Contact volunteers through the app's call interface	Service personnel provide instructions on equipment usage App product usage guide introduction interface	Premium Viewing Area / Barrier-Close Viewing Area Sign Language Interpreters Provide Real-Time Performance Translation Vibration Feedback Bracelets Pulse to Music Beats AR Lenses Display Live Subtitles and Sign Language Video Broadcasts		Visual Impaired: Access braille maps and braille translations via the app for barrier-free access to public facilities Hearing Impaired: Wear AR sunglasses to receive real-time captions of conversations around	Service personnel connect to recycling equipment	Accessible passage Accessible Bus Shuttle
Back Stage *Line of Internal Actions*	Data Data Collector Content Editing Team	Planning Specialist Identity Information Data Code	Venue Staff Information Data Code	Public Service Volunteers App Instant Messaging Technology Manual Recognition	Venue Staff Information Data Code	Site Planning Specialist Pre-recruitment and Training of Sign Language Interpreters Real-time Captioning Translation Code AR Technology Video Data Bluetooth Signal		Information Data Real-time Subtitle Translation Code	Venue Staff	Planning Specialist
Support *Line of Sketches*		The government coordinates the public transportation system		Individuals and Organizations in Social Welfare		Speech-to-Text Technology Real-Time Video Transmission Technology		Speech-to-Text Technology		The government coordinates the public transportation system
Solution										

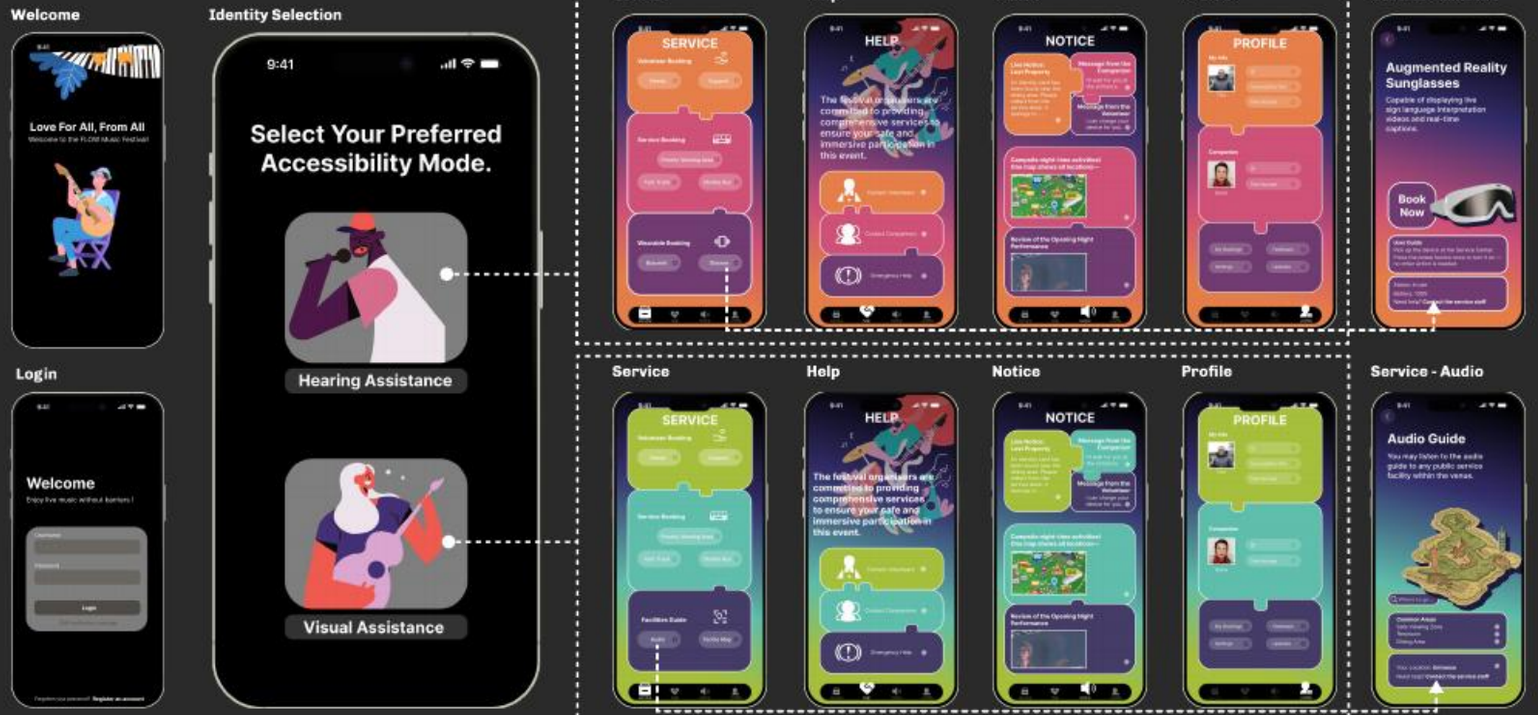
Information Map



Wireframe



Interfaces



User Testing

We blind users rely on screen readers, so the main navigation should be placed at the top.

Having the map in the notification page is confusing — it should be in the service section.



The option "I am a deaf person" feels too direct and a bit offensive; it should be phrased more gently.



Colors

- #A8BF3D
- #5FBFAC
- #453D68
- #E77F4A
- #D05278
- #703C72

Fonts

Figma Inter

Aa Medium

Aa Bold

System

IOS

Android

Sketches



3D Modelling





Outcome

Haptic bracelet



AR smart glasses





In this scenario, a DHH attendee collects the wearable device at the festival service booth, **puts on the AR smart glasses and vibration bracelet**, and opens the companion app. This setup enables him to **watch live on-stage sign language translation, receive real-time captions, and feel synchronized rhythm vibrations**, enhancing his immersive experience and engagement with the live music.

How to use



1 DHH audience pick up AR smart glasses and haptic bracelet at the service center



2 DHH audience enjoy an immersive experience during the performance



3 DHH audience can check device status via their mobile phones app